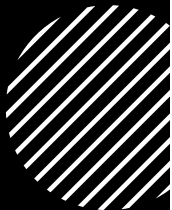


Web Scrapping in Python

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Web Scrapping



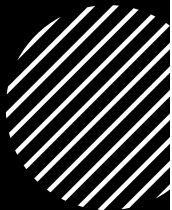
Process of **automatically extracting data from websites** using programs or scripts instead of manually copying it.

In simple terms:

A bot visits a webpage →
reads the HTML
→ extracts
required data →
stores it (CSV,
database, etc.)



Applications



AI & Machine
Learning Datasets:

Scrape
articles for
sentiment
analysis

Price Tracking &
E-commerce:

Monitor
product
prices
across
websites

Academic &
Research Work:

Extract
research
papers
metadata

Scrapping vs API

Scrapping is manual and slower, we write the code

API is automatic and faster

Always use API if available for a website access.

Exp: Twitter provides API, Paypal, Google Maps, ChatGPT, Gemini etc



Static vs Dynamic Websites



Static Websites:

- Easy to scrape using BeautifulSoup for smaller Projects and Scrapy for Large scale Projects

Dynamic Website:

- Scrapped using tools like selenium and playwright



Let's Visit.



<https://quotes.toscrape.com/>

Right click on the page and
click inspect

Install Libraries

- `pip install requests`
- `pip install beautifulsoup4`



requests – Fetching Web Pages

- Requests is used to **send HTTP requests** to websites and get their content.
- What it does:
 - Sends GET/POST requests
 - Retrieves HTML content
 - Handles headers, status codes





Example



```
import requests
```

```
url = "https://example.com"
```

```
response = requests.get(url)
```

```
print(response.status_code) # 200 means success
```

```
print(response.text)      # HTML content
```

```
--Request gets the page but doesn't understand it.
```



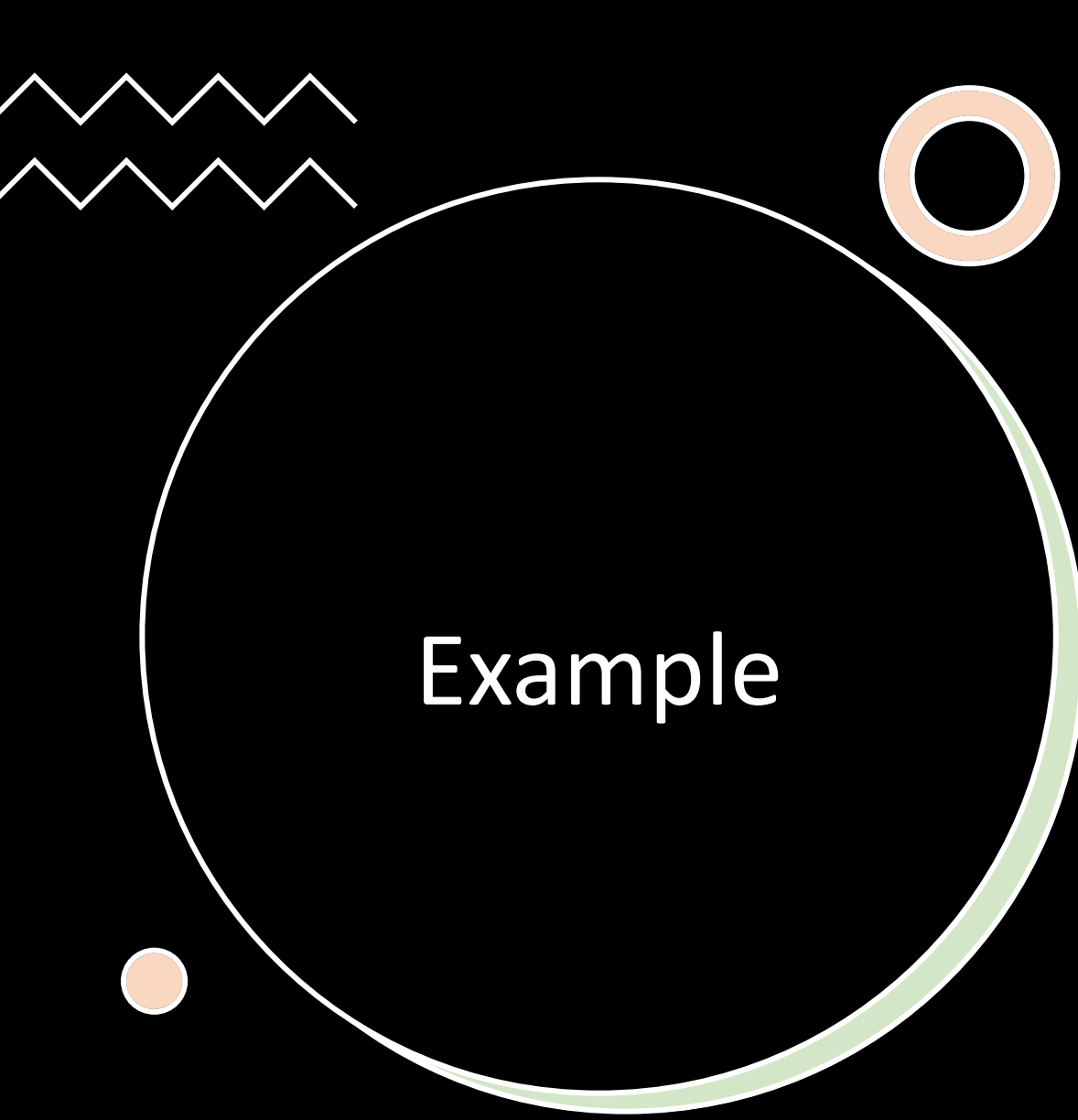
Beautiful Soup – Parsing HTML



Beautiful Soup is used to
**parse and extract data from
HTML**

What it does:

- Navigates HTML structure
- Finds tags, classes, IDs
- Extracts specific data



Example

- `from bs4 import BeautifulSoup`
`import requests`

```
url = "https://example.com"  
response = requests.get(url)
```

```
soup =  
BeautifulSoup(response.text,  
"html.parser")
```

```
print(soup.title.text)
```



Common Methods

- `soup.find("h1")` # First match
- `soup.find_all("p")` # All paragraphs
- `soup.find("div", class_="x")` # By class
- `soup.find(id="main")` # By id

Chat4Data

- Extension in Chrome
- Agentic Web Scraper, Powered by Gemini, Claude and Other Leading LLMS
- Just ask for data, it identifies pagination and give you the required data after extraction